Why APL feedback makes the effect of blocking uneven across odors

> **QUESTION**
> Why does blocking a uniform, feedback APL raise the similarity of some odor pairs more than others, and do the recordings show the same signature?

INPUTS

- The pinned model of Report 07 (uniform APL, hemibrain R and male CNS L and R; corrected calyx, new claws, new ORN table)
- The recordings (Report 06)

OUTPUTS

- `with_outside_ROI_synapses/claws_v2m2/odor_weighted/`: `TNT_feedback_vs_fixed.csv`, `TNT_feedback_vs_fixed.json`, `chance_overlap_simulation.png`, `.csv`, `.json`, `chance_overlap_data_pairs.csv`, `TNT_feedback_signature_split_half.csv`

SCRIPTS

- `scripts/tnt_feedback_vs_fixed.py`
- `scripts/chance_overlap_sim.py`
- `scripts/reports/r08_feedback_uneven.py` (this report)

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Report 07 found that a uniform APL with feedback predicts which pairs gain similarity when APL is blocked about as well as any wiring version. This report explains why, in two steps. First, feedback makes the inhibition odor-dependent, so blocking it adds the most responders to strong odors. Second, an odor that gains responders becomes more similar to every other odor by chance overlap alone.

What changed in this revision (2026-11-04)

The four inhibition rules as a table: only feedback is a model; the other three are reference rules.

New: how the Spearman values are computed, with -aPine as a worked example.

γ_o is now called the "blocked ÷ intact ratio", so it is not confused with the input gain g_o of Report 07.

In the ratio figure the data are drawn as open diamonds, so model points at the same place stay visible.

Odor strength in the data is now measured in TNT only, with the earlier mean of TNT and TNT3c kept for comparison.

Numbers come from the run with the corrected calyx, the new claws and the new ORN table.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
ϕ_o	Fraction of KCs (model) or units (data) responding to odor o	0 to 1

Symbol	What it stands for	Values, units, range
γ_o	**Blocked ÷ intact ratio** of odor o: $\phi_o^{\text{blocked}} / \phi_o^{\text{intact}}$ (in the data, TNT ÷ TNT3c; called the suppression s_o in Report 07). Not the input gain g_o of Report 07	×
CV	Coefficient of variation across odors: standard deviation ÷ mean	dimensionless
$n_a,\ n_{ab}$	Number of units responding to odor a, and to both a and b	counts
N	Number of units (or KCs)	
p_a	n_a/N, the fraction responding to a (same as ϕ_a)	0 to 1
$\sqrt{p_a p_b}$	Chance overlap: expected cosine of two unrelated 0/1 codes	0 to 1
π_k	Probability that unit k responds to a random odor (its excitability)	0 to 1
ΔS_{ab}	Change in similarity when APL is blocked	difference in cosine

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1 Step 1: feedback inhibits strong odors more

With a uniform APL, every KC gets the same inhibition aA_o for odor o (Report 05). It is the same for every KC, but not for every odor: A_o is the mean KC response, so an odor that drives many KCs recruits a lot of APL and is held down hard, and one that drives few recruits little. Feedback therefore evens out how many KCs the odors activate, like a normalization, and blocking APL undoes it. To show that feedback is what matters, three other rules for the inhibition are compared. All four start from the same pinned, blocked responses $y_{ko}^{\text{blocked}} = [x_{ko} - 1]_+$ and give the same mean fraction responding over odors (TNT3c's, 6.83%). They differ only in how the inhibition is shared out between odors (Table 2).

Table 2. The four inhibition rules. c and c_o are found by bisection in each mushroom body. In all four, the inhibition is the same for every KC for a given odor.

rule	intact response	how the inhibition is set	is it a model?
feedback (version U)	$[x_{ko} - 1 - a\,A_o]_+$, with $A_o = \frac{1}{N}\sum_k y_{ko}$	one gain a for all odors; A_o comes out of the KC responses themselves, so strong odors get more inhibition	yes: the model of Reports 05 and 07
fixed	$[x_{ko} - 1 - c]_+$	one number c for all KCs and all odors (per mushroom body): the same as raising the threshold from 1 to $1 + c$	no: a reference rule
same share	$[x_{ko} - 1 - c_o]_+$	one c_o per odor, chosen so that every odor keeps the same share of its blocked responders: $\phi_o^{\text{intact}} = \phi_o^{\text{blocked}} \times \bar{\phi}^{\text{TNT3c}} / \bar{\phi}^{\text{TNT}}$ (about 45%)	no: a reference rule
TNT3c per odor	$[x_{ko} - 1 - c_o]_+$	one c_o per odor, chosen so that $\phi_o^{\text{intact}} = \phi_o^{\text{TNT3c}}$, the data's own fraction	no: uses the answer from the data

The three extra rules are not new models of APL. They are controls that keep everything the same except how the inhibition depends on the odor: not at all (fixed), in proportion to the odor's own blocked response (same share), or exactly as in the data (TNT3c per odor). Comparing feedback with them shows whether the odor-dependence created by feedback is what matters for the fit.

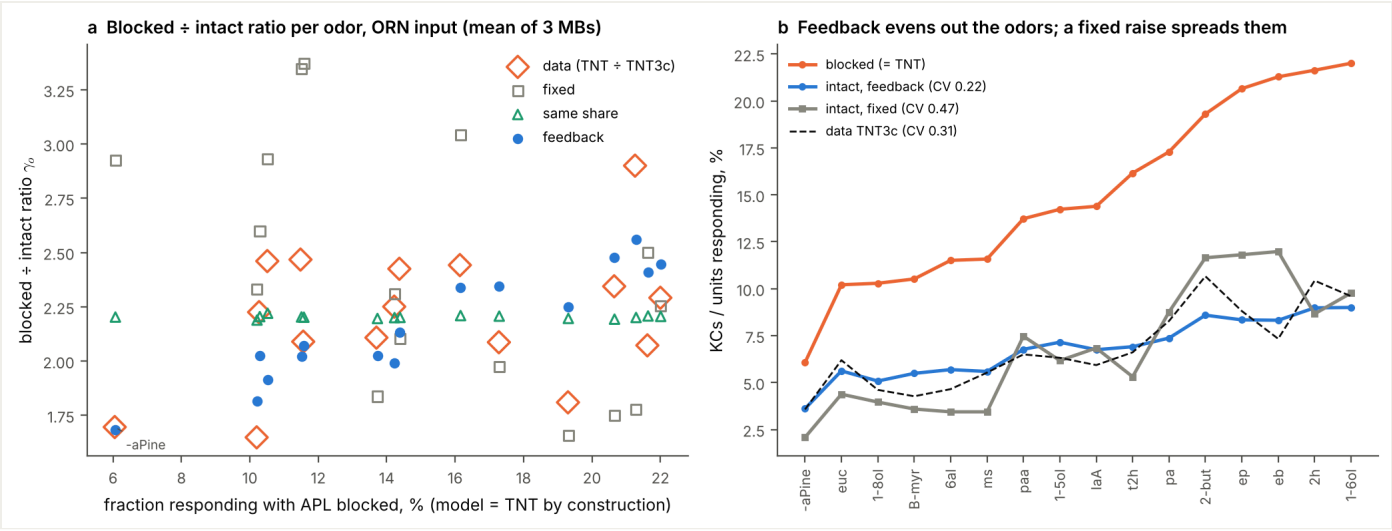


Figure 1. Feedback acts like a normalization across odors. a, Each point is one odor. Horizontal: the fraction responding with APL blocked (the model is pinned to TNT). Vertical: how many times more respond blocked than intact. Filled blue dots: feedback. Open grey squares: fixed raise. Open green triangles: same share. Large open orange diamonds: data, drawn open so that a model point at the same place stays visible inside. **b**, Per odor (sorted by the blocked fraction), the percentage responding blocked (orange), intact with feedback (blue) and intact with a fixed raise (grey), with the data's TNT3c values (dashed). CV is the spread across odors.

With feedback, the blocked ÷ intact ratio rises with odor strength (Spearman +0.92 over odors), from ×1.68 to ×2.56. A fixed raise does the opposite (Spearman -0.55): weak odors lose the largest share of their few responders, because a fixed raise removes a fixed slice of the drive distribution near threshold. In the data the ratio is barely related to the TNT fraction (Spearman +0.22).

1.1 How the Spearman values are computed: -aPine as an example

For each of the 16 odors there are two numbers, each the mean over the three mushroom bodies: the fraction responding with APL blocked (ϕ_o^{blocked}, equal to the TNT fraction by pinning) and the ratio $\gamma_o = \phi_o^{\text{blocked}} / \phi_o^{\text{intact}}$. The Spearman correlation is the Pearson correlation of their ranks over the 16 odors.

For -aPine, ϕ^{blocked} = 6.07% and ϕ^{intact} with feedback = 3.61%, so γ = 6.07 / 3.61 = **1.68** (rank 1 of 16 by ratio, rank 1 of 16 by blocked fraction). The data's value is 6.05 / 3.57 = 1.70. This γ_o is not the input gain g_o of Report 07, which for -aPine is large because its ORN responses are weak.

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2 Step 2: from more responders to more similarity

Denser codes overlap more by chance. For 0/1 responses the cosine similarity is the overlap divided by the geometric mean of the code sizes:

$$\cos(a, b) = \frac{n_{ab}}{\sqrt{n_a\, n_b}} \tag{1}$$

If odor b's responders are placed at random, independently of a's, the overlap n_{ab} follows a hypergeometric distribution with mean $n_a n_b / N$. The expected cosine is therefore

$$\mathbb{E}[\cos(a, b)] = \frac{n_a n_b / N}{\sqrt{n_a n_b}} = \sqrt{p_a\, p_b}, \qquad \text{SD} \approx \sqrt{\frac{(1 - p_a)(1 - p_b)}{N - 1}} \tag{2}$$

So an odor that gains responders becomes more similar to every other odor, and its whole row of the similarity matrix rises. This is the link between Step 1 (per-odor gains) and the per-odor part of ΔS

(Report 06, Section 4).

A floor, not the exact chance level, for KCs

The formula assumes every unit is equally likely to respond. If unit k responds with probability π_k, the expected cosine is about $(p_a p_b + \mathrm{cov}_k(\pi_a, \pi_b))/\sqrt{p_a p_b}$, which is higher when the same units respond easily to both odors. In the connectome model, two unrelated odors that each drive 10% of KCs have a cosine of 1.54 × $\sqrt{p_a p_b}$ with the ORN-table glomeruli, and 1.17 × with every glomerulus responding, because some KCs are easier to drive than others.

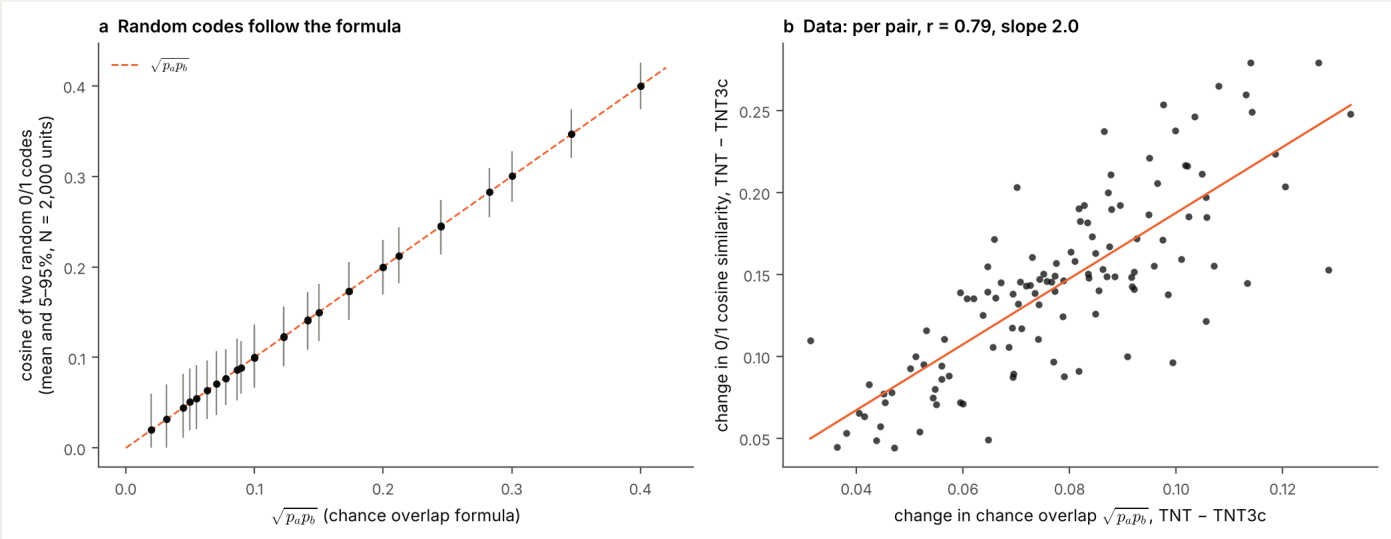


Figure 2. Chance overlap and the similarity change. a, Simulation: for pairs of random, unrelated 0/1 codes over 2,000 units, the mean cosine (dots) and its 5–95% range (grey) against $\sqrt{p_a p_b}$; the dashed line is equality. **b**, Data: each point is an odor pair. Horizontal: the change in $\sqrt{p_a p_b}$ between TNT and TNT3c, computed per recording from the fractions responding (> 1) and averaged. Vertical: the change in cosine similarity of the 0/1 responses. The orange line is the least-squares fit.

In the data the real 0/1 similarity is about twice $\sqrt{p_a p_b}$ in both conditions (0.145 vs 0.066 in TNT3c, 0.289 vs 0.144 in TNT). That excess can come from related odors or from units that respond to many odors. A shuffle that keeps each unit's number of responses and each odor's fraction reproduces the mean (0.145 in TNT3c), so with 16 odors the two cannot be told apart.

3 Step 3: which inhibition rule fits the data

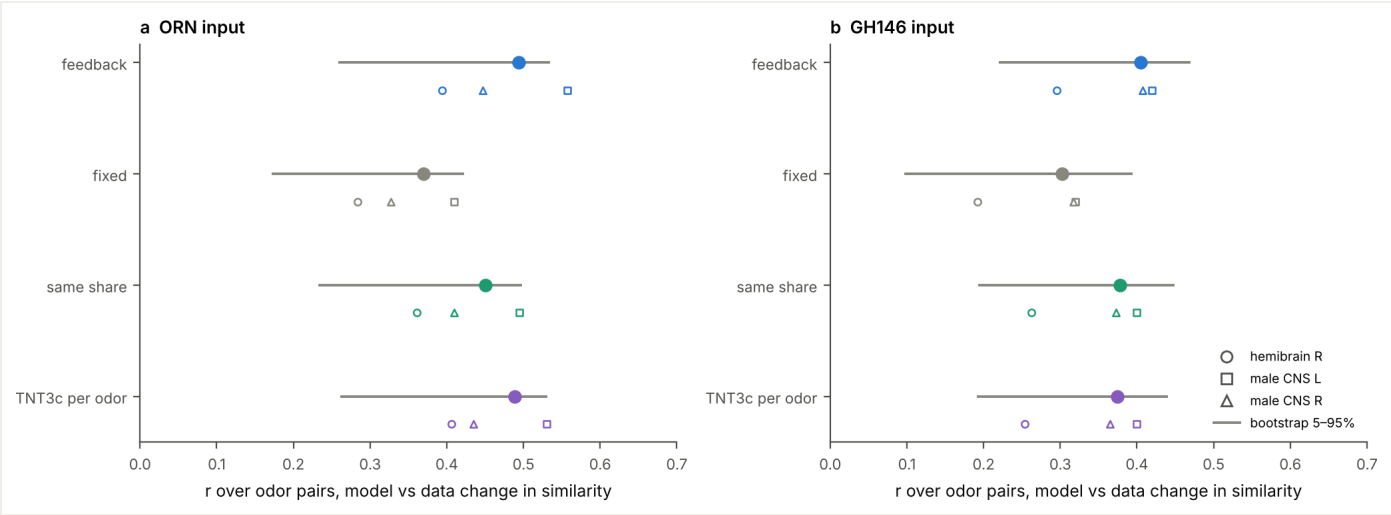


Figure 3. Fit of the four inhibition rules. For each inhibition rule, the large dot is r between the model's change in similarity (mean of the three mushroom bodies) and the data's, over odor pairs. The grey line is its bootstrap 5–95% range over recordings. The small open markers are the single mushroom bodies.

- Feedback fits better than a fixed raise in every mushroom body and in 100% of bootstrap draws (r 0.49 vs 0.37 with ORN input).
- Feedback does as well as setting each odor's intact fraction from the data (r 0.49), although it is not given the per-odor TNT3c fractions.
- The per-odor part of the change matches the data with Spearman 0.62 for feedback and 0.37 for a fixed raise.

4 Step 4: is the feedback signature in the recordings?

If APL acted as uniform feedback, odors that drive more units should gain more units when APL is blocked. Odor strength is measured in TNT, where APL is blocked: there it reflects the feed-forward drive from the PNs, and it is what the model's horizontal axis in Figure 1 uses. TNT3c is already evened out by feedback, so mixing it in would blur the measure. To avoid shared noise, strength and ratio come from separate halves of the recordings: 4 TNT recordings give the strength, and the other 4 TNT and 6 of the 11 TNT3c recordings give the ratio. This is repeated for 2,000 random splits.

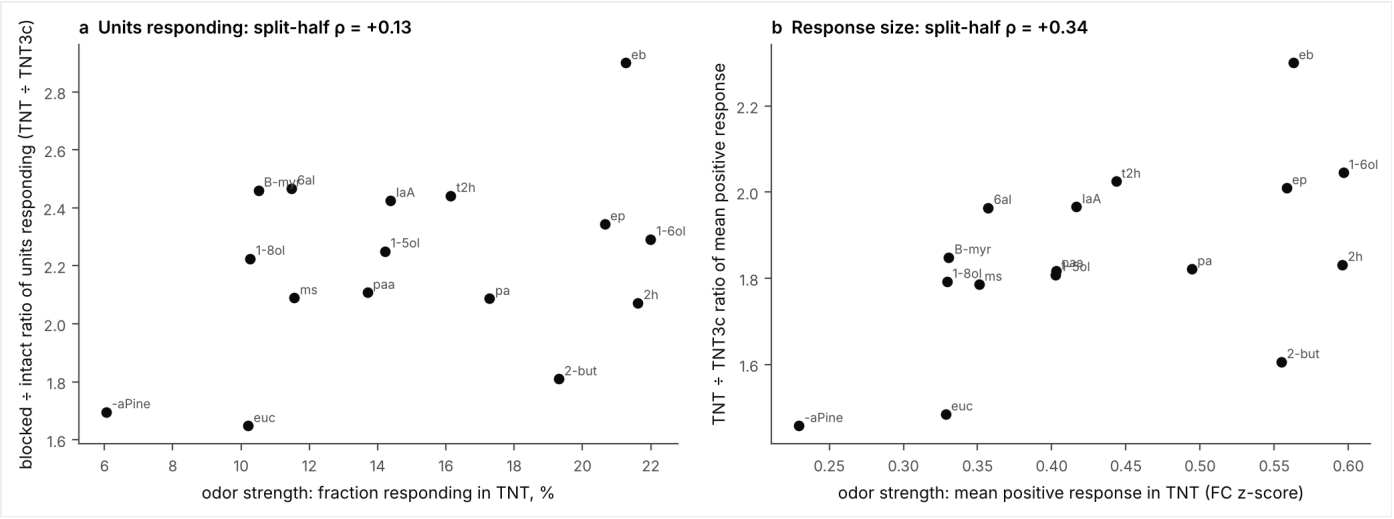


Figure 4. The feedback signature in the data. Each point is one odor, using all recordings (the split-half Spearman ρ in the titles uses separate halves for the two axes, mean of 2,000 splits). **a**, Odor strength as the fraction of units responding (> 1) in TNT against how many times more units respond in TNT than in TNT3c. **b**, Odor strength as the mean positive response in TNT (negative responses set to 0, then averaged over all units; FC z-score) against its ratio TNT ÷ TNT3c.

Table 3. The feedback signature with different measures of odor strength. Strength and ratio use the same measure in each row. The ratio always comes from the other half of the recordings. The model with uniform feedback predicts ρ = +0.92 for the fraction responding.

odor strength measured from	measure (strength and ratio)	mean Spearman ρ (2,000 splits)	5–95%	share of splits > 0
TNT only	fraction > 1	+0.13	−0.13 to +0.41	79%
mean of TNT and TNT3c (earlier version)	fraction > 1	+0.06	−0.22 to +0.33	65%
TNT3c only	fraction > 1	−0.01	−0.30 to +0.26	49%
TNT only	mean positive response	+0.34	+0.11 to +0.61	100%
mean of TNT and TNT3c (earlier version)	mean positive response	+0.27	+0.04 to +0.52	97%
TNT3c only	mean positive response	+0.21	−0.04 to +0.46	91%

- **Units responding**: the ratio is weakly related to odor strength (mean split-half Spearman +0.13, 5–95% −0.13 to +0.41, positive in 79% of splits), far from the +0.92 that uniform feedback predicts. The spread across odors is about the same in TNT and TNT3c (CV 0.32 and 0.31), whereas uniform feedback predicts less spread with APL intact.
- **Response size**: stronger odors do gain more (Spearman +0.34 for the mean positive response, positive in 99.5% of splits).
- Measuring strength from the mean of TNT and TNT3c, as in the earlier version, gives the same conclusion (Table 3).
- So the recordings show the feedback signature in response size but only weakly, if at all, in the number of responding units.

Caveats

The rules other than feedback are not mechanisms; they are reference points with the same mean inhibition.

Noise inflates TNT3c's apparent number of weak responders differently from TNT's, which can blur the gain-vs-strength relation.

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